ETABS Input Sheet, Section 6: spectrum inputs from Fig. 12

Addendum to the ETABS Elastic Modeling Input Sheet, Rev 3, Models A and B, Bangkok State Audit Office Study

Values are read from **Fig. 12 of Ornthammarath et al. (2025)**, *Far-field ground motion characteristics of the Bangkok Basin, Thailand, in the 2025 Mw 7.7 Mandalay earthquake: initial insights*, Bull. Earthquake Eng., doi 10.1007/s10518-025-02295-7. The figure shows elastic response spectra at 5 percent damping for the five basin stations and two stations outside the basin, with the DPT 1301/1302 (2018) Bangkok DBE and MCE spectra on the same axes.

How the values were obtained. Each panel was calibrated on its gridlines (0 to 10 s, 0 to 0.3 g). Every curve was then traced by its line colour, pixel column by pixel column. The digitised points were drawn back over the source figure to check them (fig_check_overlay.png). **Accuracy:** about 0.03 s in period and 0.002 to 0.003 g in Sa, which is the line width of the published figure. The code curves pass a check: the digitised DBE is two thirds of the digitised MCE at every corner, to within 0.001 g.

6.2 Function 1: DPT design spectrum, corrected to Fig. 12

The Function 1 table in the input sheet does not match the code spectrum plotted in Fig. 12. The sheet builds an ASCE-style curve from SDS = 0.15 g and SD1 = 0.20 g. That shape is not self-consistent: with Ts = 1.333 s it gives Sa(1.0 s) = 0.15 g, not the 0.20 g the sheet itself quotes as the zone 5 DBE at 1.0 s. The Bangkok basin spectrum in DPT 1301/1302 is a tabulated shape. In Fig. 12 it is a set of straight segments with corners at 0, 0.5, 1, 2, 3, 4, 5 and 6 s. Between 0 and 3.5 s the sheet's values are **21 to 40 percent below** the plotted DBE.

T (s)	Fig. 12 DBE (g)	Fig. 12 MCE (g)	Input sheet Function 1 (g)	Sheet / Fig. 12
0	0.079	0.117	0.060	0.76
0.5	0.191	0.287	0.150	0.79
1	0.198	0.298	0.150	0.76
1.5	0.182	0.274	0.134	0.74
2	0.166	0.249	0.100	0.60
2.5	0.129	0.194	0.080	0.62
3	0.093	0.139	0.067	0.72
3.5	0.073	0.110	0.057	0.78
4	0.053	0.080	0.050	0.94
5	0.037	0.055	0.040	1.08
6	0.028	0.042	0.033	1.19

Why it matters. At the building's expected cracked period of about 2.9 s, the recorded PWSA ordinate is 0.082 g. The plotted DBE is 0.100 g, so the code demand exceeds the recorded demand, which is the finding in Section 6.6. Against the sheet's Function 1 value of 0.069 g, the recorded motion would *exceed* the design spectrum at the building's period. That would reverse the Section 6.6 conclusion.

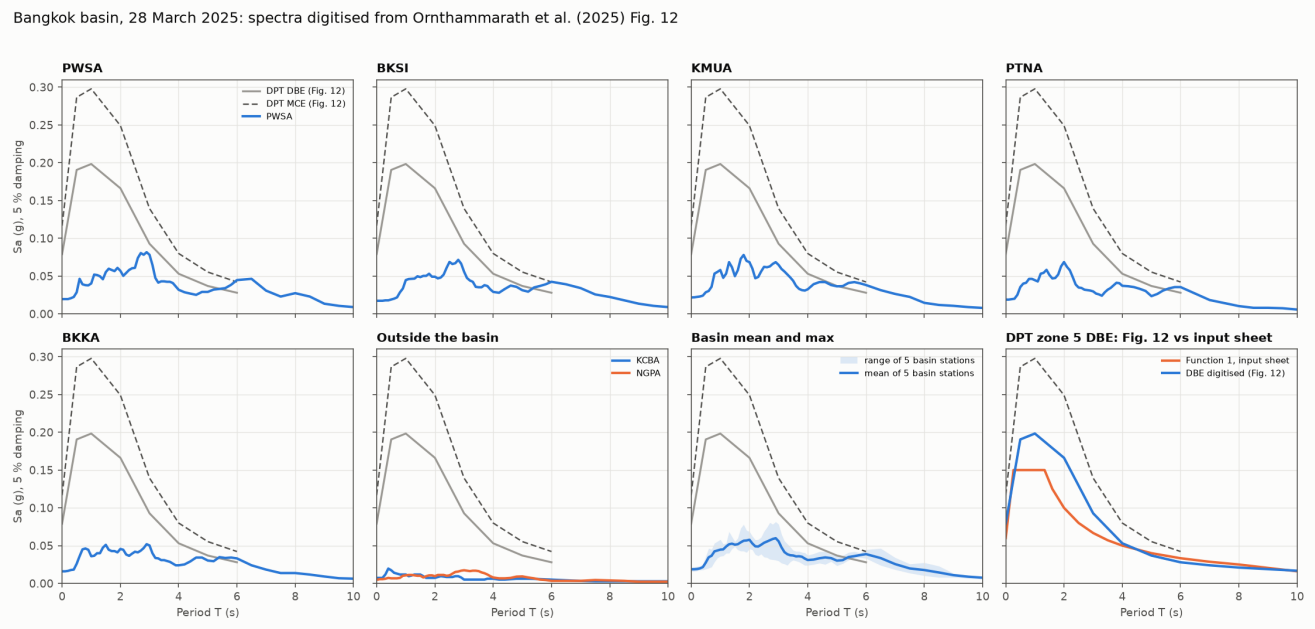
Enter this instead. Name it RS-DPT61-Z5, function type User Defined, damping ratio 0.05. These are the corner points; ETABS interpolates linearly between them, exactly as the code curve is drawn. The plotted curve stops at 6 s. Beyond 6 s, Sa = Sa(6 s) x 6 / T is **assumed** (constant velocity), so the function reaches 10 s as the sheet requires.

T (s)	DBE, Sa (g) [RS-DPT61-Z5]	MCE, Sa (g)	Source
0.000	0.0791	0.1168	Fig. 12 corner
0.500	0.1905	0.2866	Fig. 12 corner
1.000	0.1983	0.2978	Fig. 12 corner
2.000	0.1662	0.2492	Fig. 12 corner
3.000	0.0927	0.1394	Fig. 12 corner
4.000	0.0531	0.0799	Fig. 12 corner
5.000	0.0369	0.0553	Fig. 12 corner
6.000	0.0279	0.0422	Fig. 12 corner
7.000	0.0239	0.0362	assumed, Sa ~ 1/T
8.000	0.0209	0.0316	assumed, Sa ~ 1/T
10.000	0.0168	0.0253	assumed, Sa ~ 1/T

Zone check. The figure caption says only *Bangkok DBE and MCE*. The sheet identifies the curve as zone 5 because the paper quotes 0.20 g DBE at 1.0 s for zone 5, and the plotted DBE is 0.198 g at 1.0 s. If the DPT zone 5 table is available, confirm these corners against it.

2.5 percent damping (Section 6.3). If the 2.5 percent spectrum governs, the sheet gives 0.25 g DBE at 1.0 s, which is 1.25 times the 5 percent value. Uniform scaling of the table above by 1.25 is an approximation that is exact only at 1.0 s. File: RS-DPT61-Z5-DBE.txt x 1.25.

Digitised spectra



Bottom right: the input sheet's Function 1 (orange) against the DBE plotted in Fig. 12 (blue).

6.4 Function 2: 2025 recorded motion

Section 6.5 leaves the station choice open. The decision matters: at the building's period, the stations differ by a factor of about 2.5.

T (s)	DBE	PWSA	BKSI	KMUA	BKKA	PTNA	Basin mean
2.5	0.129	0.061	0.069	0.062	0.042	0.035	0.054
2.9	0.100	0.082	0.067	0.068	0.052	0.031	0.060
3	0.093	0.078	0.056	0.065	0.050	0.030	0.056
3.2	0.085	0.048	0.044	0.056	0.034	0.026	0.041
3.5	0.073	0.043	0.035	0.044	0.031	0.032	0.037

Recommendation: RS-2025-REC = PWSA, with **PTNA as a second function** for sensitivity.

PWSA is in the deepest basin and has the largest ordinates in the building's 2.9 to 3.5 s range, so it is the conservative choice. It is also the station used for the PGA and for the Phase 2 nonlinear target. PTNA is nearest to Chatuchak but sits at the basin edge, with no 3 s peak and only 0.03 g at 3 s. Running both brackets the answer for the actual site. Both use the 5 percent spectra; for 2.5 percent, PWSA is in Fig. 13 of the paper, which is not digitised here.

RS-2025-REC (PWSA), damping 0.05, Sa in g

T (s)	Sa (g)	T (s)	Sa (g)	T (s)	Sa (g)
0.000	0.0196	1.900	0.0609	3.900	0.0363
0.050	0.0196	2.000	0.0570	4.000	0.0318
0.100	0.0196	2.100	0.0503	4.200	0.0286
0.200	0.0196	2.200	0.0542	4.400	0.0268
0.300	0.0207	2.300	0.0581	4.600	0.0251
0.400	0.0223	2.400	0.0603	4.800	0.0291
0.500	0.0285	2.500	0.0609	5.000	0.0291
0.600	0.0464	2.600	0.0740	5.200	0.0321
0.700	0.0391	2.700	0.0804	5.400	0.0330
0.800	0.0380	2.800	0.0782	5.600	0.0341
0.900	0.0377	2.900	0.0816	5.800	0.0385
1.000	0.0402	3.000	0.0782	6.000	0.0447
1.100	0.0520	3.100	0.0648	6.500	0.0464
1.200	0.0514	3.200	0.0475	7.000	0.0307
1.300	0.0497	3.300	0.0413	7.500	0.0229
1.400	0.0458	3.400	0.0430	8.000	0.0274
1.500	0.0553	3.500	0.0430	8.500	0.0229
1.600	0.0598	3.600	0.0425	9.000	0.0134
1.700	0.0581	3.700	0.0425	9.500	0.0106
1.800	0.0565	3.800	0.0408	10.000	0.0089

RS-2025-PTNA (sensitivity), damping 0.05, Sa in g

T (s)	Sa (g)	T (s)	Sa (g)	T (s)	Sa (g)
0.000	0.0189	1.900	0.0629	3.900	0.0406
0.050	0.0189	2.000	0.0686	4.000	0.0371
0.100	0.0189	2.100	0.0623	4.200	0.0366
0.200	0.0194	2.200	0.0577	4.400	0.0354
0.300	0.0200	2.300	0.0486	4.600	0.0331
0.400	0.0251	2.400	0.0389	4.800	0.0303
0.500	0.0354	2.500	0.0349	5.000	0.0234
0.600	0.0400	2.600	0.0343	5.200	0.0263
0.700	0.0411	2.700	0.0320	5.400	0.0309
0.800	0.0463	2.800	0.0314	5.600	0.0337
0.900	0.0463	2.900	0.0309	5.800	0.0354
1.000	0.0446	3.000	0.0297	6.000	0.0354
1.100	0.0429	3.100	0.0269	6.500	0.0269
1.200	0.0531	3.200	0.0257	7.000	0.0183
1.300	0.0543	3.300	0.0240	7.500	0.0143
1.400	0.0583	3.400	0.0286	8.000	0.0103
1.500	0.0514	3.500	0.0320	8.500	0.0080
1.600	0.0469	3.600	0.0343	9.000	0.0080
1.700	0.0474	3.700	0.0377	9.500	0.0074
1.800	0.0520	3.800	0.0411	10.000	0.0057

BKSI, KMUA and BKKA, and the mean of the five basin stations, are in etabs_functions/ and fig12_digitized_spectra.csv on the same period grid. That grid is 0.1 s to 4 s, where the peaks are, 0.2 s to 6 s, then 0.5 s to 10 s. Below 0.3 s the spectra are flat at the PGA read from the figure: 0.016 to 0.021 g.

Entering the functions in ETABS

Step	Action
1	Define, Functions, Response Spectrum. Choose **From File**, then Add New Function.
2	Browse to etabs_functions/RS-2025-PWSA.txt (or type the table). Header lines to skip: 0. Values are: **Period vs Value**. Function damping ratio: **0.05**.
3	Name it RS-2025-REC. Click **Convert to User Defined**, so the model no longer depends on the text file.
4	Repeat for RS-2025-PTNA and RS-DPT61-Z5 (DBE file), or type the tables above.
5	Load cases as Section 7.3: RS-REC-U1 / U2 with RS-2025-REC, scale factor 9.81, no R. Add RS-PTNA-U1 / U2 the same way for the sensitivity case.
6	Check: plot each function in ETABS and compare with fig_spectra.png. The PWSA peak should be 0.082 g at 2.9 s, and the DBE 0.198 g at 1.0 s.